CODING.CARE:

GUIDEBOOKS FOR INTERSECTIONAL AI

by

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Abstract

Critical AI researchers see the urgent need to understand and rethink how AI systems are defined, developed, regulated, and mitigated. *Coding.Care: Guidebooks for Intersectional AI* argues that implementing critical approaches into AI systems more broadly requires building inviting, inclusive spaces where more people can engage creatively and critically with each other and with machine learning techniques as malleable materials. The project demonstrates craft-based, process-oriented approaches to AI that can help meet this challenge.

The dissertation presents guides for fostering critical–creative coding communities as spaces of radical belonging—activated by an ethos and a politics modeled by queer and trans* communities that embrace radical difference. This creates a basis for deep interdisciplinary thought, interrogation of formative principles, and an openness to co-creation and alternative forms necessary to reimagine AI.

With these approaches, it becomes possible to reengage emergent technologies as craftable materials, rather than unassailable forces, and to respond with impactful, sustainable, intersectional interventions. Artists, activists, scholars, and technologists can recast their relationships to emerging technologies by reframing them as crafts like crochet—deflating AI hype, lowering barriers to learning, and emphasizing sustainability and process. Thinking craft-as-technology honors the inherited knowledge of many outsider communities. Thinking technology-as-craft provides a framework to implement those theories, ethics, and tactics as intersectional critical AI.

Coding.Care collects four publications that enact the strategies it theorizes. These works address and unite a wide range of communities by relying upon different voices, forms, formats, and media. They address various stages and processes of the sociotechnical pipelines that produce algorithmic systems like generative AI — including machine learning datasets, intersectional AI, programming communities, and embodied outputs like art practice:

Coding.Care: Field Notes for Making Friends with Code gives courage to pick up unfamiliar tools, find resources to kick off a new programming project, pose questions critically, or solve problems creatively. Its pocket guide discusses how to build a cooperative, interdisciplinary community for co-learning coding like the one I have facilitated since 2019. The Intersectional AI Toolkit's co-authored zines are accessible guides to both AI and intersectionality, bringing together artists, activists, academics, makers, technologists, and anyone who wants to understand the automated systems that impact them. The work argues that established but marginalized tactics are necessary for reimagining more critical and equitable machine learning. A Critical Field Guide for Working with Machine Learning Datasets translates critical AI theories and data science concepts into practical tips for dataset stewardship. Along with the Inclusive Datasets Research Guide, it combines technical skillbuilding and critical thinking for scholars and practitioners beginning to work with large datasets, these remain the foundation of machine learning as it grows rapidly in impact. Lastly, a collection of five short lyric essays offer interludes to the major works, approaching the lived experience of an algorithmic era from oblique angles through dialogues with five 20th century artists.

Together, the guides in *Coding.Care* want to meet readers where they are—as non-academics or those bridging into new fields, looking for a common vocabulary to engage conscientiously with the urgent concerns of AI systems. *Coding.Care* offers an expansive invitation to deepen interdisciplinary conversation, apply intersectional approaches, and rework AI systems from critical–creative–caring perspectives.

Introduction

> "I'm stuck here inputting and outputting the data of a story I can't change."
>
> –Italo Calvino "The Burning of the Abominable House" (1976)

0.1 Why Coding.Care

No intervention in disproportionately harmful algorithmic systems is possible without critically aware approaches to technologies from deeply plural perspectives. Meanwhile, no such proliferation of perspectives is possible without inviting spaces to understand, interrogate, and reimagine the infrastructures that support those systems. *Coding.Care: Guidebooks for Intersectional AI* argues for the essential entanglement of critical AI approaches and creative-critical coding communities, showing how each needs the other. It shows what intersectional, interdisciplinary, creative–critical approaches to AI systems and other emergent technologies can look like. Its multimodal guides apply these approaches as in-practice experiments — in different contexts, for different audiences, for different aspects of these urgent issues.

The stakes are high: technologies like machine learning urgently require transformative interventions that recalibrate these systems' values and their stakeholders. Automated decision-making systems disproportionately harm the marginalized majority. The communities most impacted by them, and best poised to intervene, currently go unheard as powerful actors profit from their data and labor.

Large companies warn of hyperbolic coming dangers in order to distract from the current dangers they perpetuate (Kapoor & Narayanan, 2023). Already-toothless policy recommendations are watered down and ignored (Heikkilä & Ryan-Mosley, 2023). Meanwhile more and more data represent less and less diversity (Bender et al., 2021), more and more processing power destroys more and more planet (Dodge et al., 2022). Glib fun with AI on one side of the world relies on extractive data and labor for pennies a day on the other (*Exclusive*, n.d.; Nedden & Dongus, 2017). But why must it be this way?

> "We must begin with the knowledge that new technologies will not simply redistribute power equitably within already established hierarchies of difference. The idea that they will is the stuff of utopian naivete and technological determinism that got us here to begin with." (Sharma, 2020)

We cannot expect technology to solve the problems of technology. But with critical engagement, we as users, makers, scholars, and arbiters of tech can reclaim more collective agency and access by considering tech at tangible scales while considering its systemic impacts. We can transform technologies themselves rather than accepting their current shapes.

How do we participate on our own terms, in order to change systems to suit our communities–especially when learning programming and engaging with tech is often so intimidating? How do we keep from further marginalizing those whose perspectives are most necessary in order to face the challenges of emerging technologies and to change them? How do we reclaim technology as a human craft?

> "Anyone who has ever woven or knitted knows that one can change patterns […] but, more importantly, they know *there are other patterns.* The web of technology can indeed be woven differently, but even to discuss such intentional changes of pattern requires an examination

of the features of the current pattern and an understanding of the origins and the purpose of the present design.” (Franklin, 2004)

I argue that such change requires fostering critical–creative coding communities as spaces of radical belonging, activated by an ethos and a politics modeled by queer and trans* communities that embrace radical difference. This creates a basis for deep interdisciplinary thought, interrogation of formative principles, and an openness to co-creation and alternative forms necessary to reimagine AI. With these approaches, it becomes possible to reengage emergent technologies as craftable materials, rather than unassailable forces, and to respond with impactful, sustainable, intersectional interventions.

My interest in code and AI systems grew from wanting to understand how I might write books in new forms, using digital tools that I wanted to build. My background shapes my stakes in these urgent concerns and informs how I understand them.

Perhaps like many of you, I came late to both queerness and coding.

I spent a long time struggling to learn in tech (and other) communities where I felt I didn’t belong. Still, I was conscious of the growing impacts that digital tools were having on my voice and on others who might also feel left out.

Over the last decade, I have begun developing the approaches shared across the works collected in *Coding.Care*, as practices of exploration and investigation, experimentation and imagination, repair and resistance toward and with technologies like machine learning. These are practices that ask how to better use technology toward relationality and building the systems and worlds we want.

0.1.1 AI Needs Critical and Intersectional Approaches

A large interdisciplinary community is pushing for understanding, critiquing, and rethinking how we define, develop, deploy, regulate, use, and mitigate the effects of machine learning tasks, datasets, models, algorithms, architectures, and agents, which we collectively and nebulously understand as ‘artificial intelligence' or 'AI' systems. The field of critical AI includes detailed and varied analysis of the pitfalls of existing methods, which cannot be addressed with technical improvements alone (Raley & Rhee, 2023). In this way, it is also distinguished from applied approaches like 'AI for Good' or 'AI for Society', which can lack critical perspectives despite their intended altruism. Critical AI research pairs with urgent calls for alternative Indigenous (*CARE Principles of Indigenous Data Governance*, n.d.; "INDIGENOUS AI," n.d.), antiracist (*Anti-Racist HCI: Notes on an Emerging Critical Technical Practice | Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems*, n.d.), anticolonialist (Chakravartty & Mills, 2018; Rights, 2020), crip (Hamraie & Fritsch, 2019), neurodivergent (Goodman, n.d.), queer (Keeling, -01-22 2014; Klipphahn-Karge & Koster, n.d.), intersectional (Ciston, 2019; Klumbytė et al., 2022), and other knowledge systems to be applied to machine learning and other emergent technologies. Yet, importantly, none of these mixed methods of analysis and intervention has yet to be adopted widely into standard machine learning practices, even as the use and awareness of AI escalates and its issues grow more urgent.

Industry implementations of so-called ethical tech reduce complex concepts into flattened ideas of fairness and representation (Ovalle et al., 2023). Machine learning, as a mass production, produces a false sense of certainty from uncertainty (from millions or billions of small uncertainties) argues political geographer Louise Amoore, describing its reductionist logic(Amoore, 2020). These uncertainties are also claims about "what we should know, how we should know what we know, and how that knowledge should be deployed. Each exposure to a dataset occurs because someone concluded that the information in that dataset should be used to determine a possible future" (Hakopian, 2022).

While the need for AI oversight is clear, many are calling for total overhaul, like computer scientist Joy Buolamwini (Buolamwini, n.d.) who argues for what she calls algorithmic justice. Overhaul and even oversight can appear out of reach when the scale of AI systems seems unfathomable and entangled. The valid criticism that algorithmic systems are biased because their data are biased — often summed up “garbage in, garbage out” — sets up a quest already doomed to fail. Yes, in many cases it would be nice to have more, better data. But what would be better data? Or an optimized system? For what goal, and for whom exactly? There is no such thing as unbiased. There is only “good enough” data — and only sometimes, for some tasks. As [xxx-id] Yanni Alexander Loukissas puts it, “all data are local,” meaning they come from particular contexts and are shaped into ‘data’ by the processes which produce them into datasets (Loukissas, 2019). There are many ways to do any task, informed by minute choices at every step. As these choices scale exponentially with computation, their impacts magnify exponentially too.

For example, because generative AI models have become so large, and training processes so expensive, they frequently rely on foundation models: These are previous models, often designed for other ‘general’ tasks, used as the building blocks for new models. Like a sourdough starter, foundation models carry with them the histories of how their datasets were designed and for what purpose, whose data was included or excluded, and the choices their creators made when preprocessing them. These are often decades-old “benchmarks” that leave traces of debunked or erroneous information in the outputs of so-called cutting-edge models. Their errors are compounded by the computational speeds that allow thousands of operations to run per second. Yet, because they are processed through algorithmic systems, automation and AI hype helps normalize, naturalize, and neutralize these very subjective decisions. Artist Hito Steyerl argues, “the supposed elimination of bias within datasets creates more problems than it solves. The process limits changes to parts of the output, making these more palatable for Western liberal consumers, while leaving the structure of the industry and its modes of production intact” (Steyerl, 2023). Thus, “fixing” inputs or outputs does little to address the structures of the systems (both cultural and technological) which produced them. When the stakes are too high, no data could be good enough to make life or death decisions. We cannot rely on computational systems for infallibility and rationality, nor can we look to these technologies uncritically as bandaids for the problems they exacerbate.

And yet, practical applications of these valid critiques and calls for change remain difficult to implement. How do we get there? We need a different approach.

As I have previously theorized it, intersectional AI calls for demystifying normative AI systems and learning from marginalized ethics and tactics, in order to fundamentally transform AI. It requires multimodal, polyvocal, experimental approaches that cut through the technological solutionism (Ciston, 2019). It requires slow, long-term investments in algorithmic justice, rather than extractive, performative forms of inclusion that erase friction, context, and agency as they scale up for machine learning tasks (Sloane et al., 2022). Critical AI researchers and makers engage these systems as sociotechncial objects embedded in their historical, social context, say critical AI researchers and professors Rita Raley and Jennifer Rhee. They argue that we must be "situated in proximity to the thing itself, cultivating some degree of participatory and embodied expertise, whether archival, ethnographic, or applied (Raley & Rhee, 2023). This level of engagement requires interdisciplinary and intersectional perspectives in order to permeate the entire AI pipeline, transforming it altogether.

Intersectionality, Kimberlé Crenshaw's iconic analysis of institutional power is often misinterpreted as individual identity politics (Crenshaw, 1989). In fact, intersectionality "critiques systems of power and how those systems structure themselves to impact groups and individuals unequally" (Cooper, 2016). It is as useful for understanding privilege as it is for understanding marginality (Crenshaw, 2021). Intersectionality can reveal the tangible human and more-than-human costs entangled in these algorithmic systems – from proliferating data and its material infrastructures, to consolidated power and its sociocultural infrastructures. Intersectional analysis shows that such power is differential by design. Conversations about AI fairness, transparency, explainability, ethics, public good, and the hype cycles of new technologies are grossly incomplete without intersectional analyses of power and intersectional tactics of (and beyond) equity and inclusion. No change *about* us *without* us.

Throughout this project, I adopt the term 'trans*formative' because it looks for root causes and radical alternatives, and suggests queer and trans* imaginaries. Here this means completely reworked alternatives to the computational logics that perpetuate harmful systemic inequities. Rhetorician Adam Banks argues for what he calls "transformative access" to digital technologies(Banks, 2006). He says that access is more than owning or using tools, participating in processes or even critiquing their failings. Transformative access is "always an attempt to *both* change the interfaces [where people use that system] and fundamentally change the codes that determine how that system works." Banks highlights the important role Black people play in technology's transformations, saying, "Black people have hacked or jacked access to *and* transformed the technologies of American life to serve the needs of Black people and all of its citizens" (Banks, 2006). As many Black feminist and intersectional theories argue, supporting the needs of people who are pushed to the margins often leads to more effective support for many others as well. Access and agency with technology is not a favor or a handout; more encounters across communities with different backgrounds and knowledge, engaging and imagining different possibilities for technology benefits everyone.

Hyped AI discourse explores limited questions about AI because it continues to draw from limited perspectives, letting normalized narratives about humans and automated systems frame the terms of debate. "Machine learning [...] is an expression of that which has already been categorized," says digital culture researcher Ramon Amaro (2022). Not to think to train on a variety of faces, or not to raise concerns about training on faces at all, happens because there is not a variety of perspectives in the room when these very human decisions are being made: before, during, and after the data is being collected; and before, during, and after the code is being written and run. The spaces where technologies are discussed, designed, and implemented are missing essential perspectives of those pushed to the margins, who are most capable of addressing the concerns facing technology now. These concerns are not new, nor strictly digital. Such questions — about representation, equity, ethics, and more — have been addressed by a wide range of communities with different types of knowledge for centuries. Yet intimidating, isolating cultures around the specialization of computation and programming practices have left so many people out of these conversations.

With the goal to imagine different systems, code literacy should not be defined only on the narrow terms of those creating existing systems (Vee, 2017). It is clear that bootcamps and hiring initiatives, though useful, do not support the goal of shifting a variety of voices into positions where can effectively make change (Abbate, 2021; Dunbar-Hester, 2019; Hicks, 2021; Vee, 2017). First, joining an 'elite' tech field is a moving target, entangled with race, gender, and economic politics that rig the game. Communications scholar Christina Dunbar-Hester and others call for interventions that go deeper than training more people in tech jobs, pointing out that this does little to examine the structures that organize and value work sectors. She argues this calls for "a larger reevaluation and appropriation of categories themselves—the boundaries of what is 'social' and what is 'technical' are flexible categories" (Dunbar-Hester, 2019). Second, diversity quick-fixes do not acknowledge the many people already participating in the production of technologies in the global majority, from those harvesting of rare earth minerals and circuit board manufacturers (AI Now Institute, n.d.; Nakamura, 2014) to the content moderators (Roberts, 2016) to the crowd workers (Sunder 2022). Part of reworking inclusion involves acknowledging how many more people are already engaged with sociotechnical practices and impacted by them, as user-practitioners, data subjects and subjectees (Ciston, 2023), skilled crafters and critics. It requires rethinking access as mutually beneficial connection across communities of practice, by finding shared spaces and common vocabularies.

0.1.2 Critical, Intersectional AI Needs Creative, Care-full Approaches

How do we reconnect the communities of practice who are currently building technologies and those who are equipped with the knowledge necessary to consider its most urgent questions?

Coding.Care argues for craft-based, process-oriented, community-driven approaches to AI that can better help meet this challenge. It demonstrates that implementing critical approaches into AI systems more broadly requires building inviting, inclusive spaces where more people can engage both creatively and critically with each other and with machine learning techniques as malleable materials.

As important as the criticisms of current AI systems are, effective critique should be imbedded and actively practiced in order to produce alternatives. Abstract calls for access are not enough; if we want the means to imagine and create truly transformative alternatives, we need spaces that acknowledge our diverse capacities and connect us in shared goals. Caring, creative, and critical approaches must be combined in order to adapt these conversations and these technologies to welcome different communities and the wider range of knowledge necessary for such transformation.

Critical methods activate creative modes and root them in sociotechnical complexity. Creative methods in turn activate critical modes and root them in care and connection, taking the critical out of the abstract and into action. Strategies for combining creative and critical practices have found disciplinary grounding around the world, whether termed research–creation, arts-based research, artistic research, or design research (Fournier, 2021; Loveless, 2019; Willis, 2016). Importantly, critical–creative is not a binary but a blending that is needed to address AI's urgent concerns.

'Critical–creative coding' is the term I use to describe combining the application of critical AI approaches with creative coding, an existing set of artistic approaches to software and the diverse communities surrounding these approaches. Creative coding has long lineages that can be traced to 1990s net art, analog fine arts, and early computation. Like 'tactical media' (Raley, 2009) and 'critical engineering' (Oliver et al., 2011), critical–creative coding creates software and other technological objects not only for their aesthetic qualities but in order to investigate them as objects of study and critique. Like 'critical making' (Bogers & Chiappini, 2019) and many 'makerspaces' (Dunbar-Hester, 2019), critical–creative coding considers the tools it takes up and the community practices which root it.

Inspired by all of these practices, critical–creative coding consolidates sets of practices I have witnessed in all of these communities and others. Building on these inspirations, [I have found in my own practices and community-building that it is key that] critical–creative coding also emphasizes co-learning and process-oriented practices from crafting communities, and an ethos and a politics of radical belonging modeled by queer and trans* communities that prioritize radical difference.

Process-oriented practice (creativity, craft) and radical belonging (care) are at the core of this approach. Care and creativity are not additions, affectations, or antonyms to critical theory or technical savvy; rather, they are its central fortifications. The technical means (coding skills), the analytical means (analytical, political, aesthetic, ethical contexts), and the material means (data, energy, hardware, infrastructure, care) are inseparable. Code work is critical work is care work is creative work.

The emphasis on craft and creativity is not exclusively aesthetic but also strategic. Artists, activists, scholars, and technologists can recast their relationships to emerging technologies by reframing them as crafts like crochet. These strategies help deflate AI hype, lower barriers to learning, and emphasize sustainability and process. Thinking craft-as-technology honors the inherited knowledge of many outsider communities. Thinking technology-as-craft provides a framework to implement those theories, ethics, and tactics as intersectional critical AI.

> "Questions like 'is a computer creative' or 'is a computer an artist' or the like should not be considered serious questions, period. In the light of the problems we are facing at the end of the 20th century, those are irrelevant questions. Computers can and should be used in art in order to draw attention to new circumstances and connections and to forget 'art'." (Nake, 1971)

Questions about art and creativity in relation to AI are bigger than beauty; they are fundamentally questions trying to define humanity: *What makes us creative or empathetic? What makes us different from machines? What makes us human?* They are old, old questions. They emerge from centuries of colonizer thinking that frames 'man' as an idealized, individualized white subject. These questions must be addressed with creative–critical approaches that open up space for play and experimentation beyond these constraints. Combined with critical tools, artistic experimentation is a powerful method for worldbuilding that can challenge deep-seeded paradigms in machine learning, data science, and technology communities — and can formulate new ones with diverse perspectives. As an intervention into deep learning algorithms, designer and researcher Catherine Griffiths argues for `reflexive software development` that critically considers and interactively presents the circumstances of its own production (Griffiths, 2022). The outputs of such research can be tools that continue to probe their research questions, both through the very processes of their creation and through their later use by others. Cinematic arts professor Holly Willis argues that "arts-based research is rooted in critical theory, framing the research process within the context of power, emancipation and a deep questioning of the ethical and ideological implications of knowledge and change" (Willis, 2016). For me, artistic research has the capacity to combine rigorous scholarly investigation, deep community building and activism, and creative play in ways that facilitate connections across broader non-academic audiences.

> "in failing to fully belong, and allowing that nonbelonging to denaturalize, emergently, its givens, research-creation tells other stories, uncanny stories, that (have the potential to) carry within them [...] other ethics" (Loveless, 2019).

Queerness and art have been two key radical practices for me. Art has been the space where I am able to unpack complex ideas for myself, because I can treat them more freely as artistic materials. It is where I am able to follow instinct and feel into how my tools, platforms, and forms shape their outputs and outcomes. It is also the space where I feel able to imagine wildly, creating digital objects that should exist but don't, or couldn't exist but might.

Artistic research opens space beyond research questions, where research tensions live. In that space, I can sit a bit longer with questions I know I cannot answer, questions that make me uneasy. I can hold two contradictory ideas simultaneously and let them push–pull me, forgiving myself imperatives and outcomes, productivity and proven hypotheses. I can put myself into the trouble, because I already embody these questions in my lived experience.

This imaginative work is central to supporting very practical next steps and strategies. It is central to supporting more access to communities of practice where others can continue the kinds of artistic experimentation that challenge paradigms and creates new forms I might never imagine otherwise.

[xxx][more about queerness/less about practice?]

Code can [be/mean] so much more — [if we let it]. Code is collaborative, says Mark C. Marino, who helped develop the practice of Critical Code Studies[, which reads software code with the close attention as literature]. Code can be an inviting, interpretive practice: "Code's meaning is communal, subjective, opened through discussion and mutual inquiry, to be contested and questioned, requiring creativity and interdisciplinarity, enriched through the variety of its readers and their backgrounds, intellectually and culturally" (Marino, 2020). I argue that such approaches invite the creation of hybrid communities that acknowledge the interdisciplinary capacities of programming, and the diverse capacities for knowledge.

I formulate this approach and its goals as 'crafting queer trans*formative systems'. It requires [space/access], care, queerness, radical belonging. Process-oriented, craft-focused practice and radical belonging make room for in-betweenness — for rejoining the divisions between theory and practice, between user and programmer, which were artificially split from the start (Artist, n.d.; Hoff, 2022; Nardi, 1993); for un-siloing domains and disciplines, the artificial boundaries that divide technologists from activists and critics from creators; for finding common language and common values that come with working knowledge of whole systems. They let everyone's contributions be valued, and make efforts toward understanding mutual, because all participants realize that we each make important contributions.

"any theory that cannot be shared in everyday conversation cannot be used to educate the public." (hooks, 1994)

0.2 What's in This Collection

Each section of *Coding.Care* puts its thinking into action — tackling related aspects in different forms and for different audiences. The parts combine to enact the ethics and tactics described in this introduction. Together these public-facing resources provide plainspoken translations of technical, critical, aesthetic, and ethical concepts relevant to technology. They prioritize finding a common language to connect across boundaries. They are written as jargon-free as possible to allow them to travel as broadly as possible. They also vary in mode and methodology. They are produced in approachable, non-academic formats like zines, and in contexts like public workshops, in order to support discussions across communities of practice — including code creators, AI researchers, and marginalized outsiders. Together, they show how different ways of knowing are necessary to address the same questions raised by automated technologies, as well as how different aspects of these urgent issues must be addressed simultaneously.

0.2.1 TRANS: *Crafting Queer Trans*formative Systems, an Introduction*

[XXX][TO-DO]

As an extended introduction to the theories, metaphors, and tactics applied throughout this collection, *Crafting Queer Trans*formative Systems* looks at [XXX]

In *Crafting Queer Trans*formative Systems* I go into the [XXX] behind the need for new [XXX], what makes a system [XXX], and how to imagine and build them. It details the principles I draw from crafting, from queer and trans* community-building, and from coding in the arts to

0.2.2 CRAFT: *Coding.Care: Field Notes for Making Friends with Code*

Coding.Care: Field Notes for Making Friends with Code is a pocket guide to sustaining friendly coding communities — why we need them, how to build them, how to let them thrive. It focuses on lessons I learned from Code Collective, the diverse hack lab that I started in 2019 when I yearned for the adaptable, encouraging environment I had needed when I was first struggling to learn to program. In gratitude to teachers like Brett Stalbaum, at UC San Diego's Computing in the Arts program, who had showed me code could feel creative instead of prescriptive, I wanted to make a space where I wouldn't feel like an outsider for 'not knowing everything' about programming, and I suspected others might feel the same.I wondered how to recreate that experience.

In Code Collective, a mix of media artists, activists, makers, scientists, scholars, and engineers gather to co-work and co-learn, thinking critically with code in an inclusive, interdisciplinary space that supports many kinds of learners. The Collective unites students who may have zero technical experience with those who may have lots of technical experience but perhaps lack a critical or creative lens; and we value their experiences equally, reinforcing the idea: **"We all have something to teach each other."**

This guide looks at a variety of the strategies and tools we have explored and developed as we have grown. It discusses how we have adapted to meet the needs of our community — from hosted workshops to hybrid-format meetups, from pandemic support to alumni programming. Code Collective's approaches draw on many existing methodologies and methods from intersectional queer, feminist, anti-ableist, and anti-racist theories. The guide connects these approaches to cooperative organizations Varia and p5.js, and to Critical Code Studies and to practices like working iteratively and breaking critically.

As a guide for making friends with code, *Coding.Care* discusses how practices such as process-oriented skillbuilding, co-teaching and co-learning, and snacks (always snacks) embody the Collective's guiding values, such as **"scrappy artistic strategies not perfect code."** The guide shares projects and feedback from members of the Collective, who report how these values and practices have shaped them as emerging makers and thinkers. Personally, I have found this community to be the strongest influence on my own research, above and beyond my role as facilitator. Code Collective has become a joyful space for creative risk-taking that nourishes my practice. The guide offers practical advice for getting comfortable with code, while situating these approaches and groups within an **ethics of coding care** — grounded in shared embodied knowledge, embedded co-creation, and programming with and for community — as an antidote to technocratic values and as an enactment of its ethos.

In her book, *Coding Literacy*, Annette Vee argues that, "Changing 'how the system works' would move beyond material access to education and into a critical examination of the values and ideologies embedded in that education. [...] Programming is a literacy practice with many applications beyond a profession defined by a limited set of values" (Vee, 2017). Vee calls this kind of programming access "transformative." Through *Coding.Care*'s intimidation-free, learner-led, process-oriented approaches, it both theorizes and models the creation of caring communities and innovative spaces that can transfer knowledge across social strata and intellectual disciplines in order to reshape technological systems.

OBJECTIVES: Through *Coding.Care*, understand how to approach programming with less fear and more fun, with less constraint and more community support. Think creatively and critically about the kinds of technologies you want to make and support. Learn to choose and use tools, languages, and platforms that match your goals and ethics. Create or join communities of practice that feel supportive and generative.

0.2.3 FORMATIVE: *A Critical Field Guide for Working with Machine Learning Datasets* and *Inclusive Datasets Research Guide*

Datasets provide the foundation for all of the large-scale machine learning systems we encounter today, and they are increasingly part of many other research fields and daily life. Many technical guides exist for learning to work with datasets, and much scholarship has emerged to study datasets critically (Corry et al., n.d.; Gillespie & Seaver, 2015). Yet no guides attempt to combine technical and critical approaches comprehensively. Every dataset is partial, imperfect, and historically and socially contingent — yet the abundance of [problematic datasets and models] shows how little attention is given to these critical concerns in typical use.

A Critical Field Guide for Working with Machine Learning Datasets helps navigate the complexity of working with giant datasets. Its accessible tone and zero assumed knowledge support direct use by practitioners of all stripes — activists, artists, journalists, scholars, students — anyone who is interacting with datasets in the wild and wants to use them in their work, while being mindful of their impacts. Developed with Kate Crawford and Mike Ananny, as part of their research team Knowing Machines, the field guide discusses parts and types of datasets, how they are transformed, why bias cannot be eliminated, and questions to ask at every stage of the dataset lifecycle. Importantly, it shares benefits of working critically with datasets when (on the surface) it may seem just as easy not to.

The *Inclusive Datasets Research Guide* is an interactive digital guide for academic researchers working with datasets, that supports them with an overview of key concepts and considerations for working with datasets, as well as providing tools and software, books and tutorials, and recommendations for thinking inclusively. Like the Critical Field Guide, the Inclusive Datasets Research Guide focuses on a blend of technical and critical decisions that arise when working with datasets. Because this guide is aimed at students and teachers, the format is brief collections of resources rather than conceptual deep-dives. The guide appears on the USC Libraries' website along with its other research guides on many topics.

Developed by a team at USC Libraries, with the support of a grant from the USC Office of Research, this research guide was written as part of a grant to acquire core research datasets to support areas of inquiry by USC researchers into arts, humanities, and machine learning. I was recruited to provide interdisciplinary perspective on inclusive approaches to machine learning, and I joined a team including a chief library technologist, data science graduate students, special collections librarians, a research communications specialist, and a multimedia digital humanities specialist. We conducted 18 interviews with faculty across campus who worked with datasets in order to develop an internal rubric to support collection development. Through this process, we found that the need was less for dataset acquisition, because researchers did not look to libraries for their datasets but of course had access to many elsewhere. Still, the rubric we developed was utilized to acquire approximately 50 collections identified for being more accessible, inclusive, 'datafiable', and meaningfully engaged. Instead, we found the need existed for more curated resources and more training on how to select and use datasets critically, while remaining mindful of their origins and impacts — which led to the expanded aim of the grant and the development of the *Inclusive Datasets Research Guide*

Both the Critical Field Guide and the Inclusive Datasets Research Guide reflect on the stakes of datasets and the human choices they relies on. Reframing the information in two different forms shows that it can be more effective in different contexts, depending on the audiences and the rhetorical tone they require. Both works are examples of how concepts and processes researched in the *Intersectional AI Toolkit* (below) can be reworked for new institutional contexts. Adapting the Toolkit to new audiences in library science, data science, and the social sciences posed interesting challenges that both expanded and refined the work. It required the ideas be scaled up and applied, and sometimes renegotiated until their rewordings no longer felt like watering down. So much of [the work] I am still learning, is about [encountering people where they are][which means remembering that I have as much or more to learn from others than the other way around.] In each project, I got to learn how another field has addressed the problems of knowledge organization and bias, historically and in the present. Library scientists, of course, have at least a century of practice considering questions of how to categorize, curate, and archive. Social scientists have been asking how and what to measure for just as long. None of this is perfect, either, but learning from each institution, and combining thse with what machine learning is trying to ask, helps me understand better how we got where we are today. Combining these with what other intersectional practices may already understand helps me understand better what each domain might learn from the other.

OBJECTIVES: Through the *Critical Field Guide for Working with Machine Learning Datasets* and the *Inclusive Datasets Research Guide*, understand the importance of working critically with datasets as part of any machine learning practice. Identify the parts, types, and functions of datasets as you encounter them. Determine whether a particular dataset is a good fit for your project by asking understanding critical questions to ask at each phase of the dataset lifecycle. [Work with the communities impacted by your research to create strategies for addressing potential harms in the datasets you utilize.]

0.2.4 SYSTEMS: *The Intersectional AI Toolkit*

The *Intersectional AI Toolkit* argues that anyone should be able to understand AI and help shape its futures. Through collaborative sine-making workshops, it aims to find common vocabularies to connect diverse communities around AI's urgent questions. It clarifies, without math or jargon, the inner workings of AI systems and the ways in which they operate always as sociotechnical systems. The Toolkit celebrates intersectional work done by many other researchers and artists working to address these issues in interdisciplinary fields; and it gathers and synthesizes legacies of anti-racist, queer, transfeminist, neurodiverse, anti-ableist theories, ethics, and tactics that can contribute valuable perspective. Its three formats allow for multiple entry points: The digital wiki offers a forum for others to discuss and expand upon its topics. The collection of printed zines share AI topics at a concise, approachable scale. And the in-person and hybrid-online workshops invite multiple communities to participate directly in the systems that impact them.

Selecting the toolkit format was a key consideration of the development process for the *Intersectional AI Toolkit*. The toolkit form taken up here was first modeled after Ahmed's 'Killjoy Survival Kit' (Ahmed, 2017).[1]. The term 'toolkit' was thoroughly contested — too instrumental and object-oriented — but settled upon after nothing else quite suited. Not compendium or catalogue or care package, not index or hub or gazetteer, not manifesto or knapsack or portal. The technologies used went through many iterations, from git repo to wiki to self-hosted hybrid back to repo again, in search of a platform that would facilitate guest user access without heavy onboarding, track edits, and adapt to multimedia zine forms. I am still remaking the work and searching for the perfect form. I suspect I will have to create it, and it will continue to change. In its various iterations, the Toolkit has grown into eleven zine-making workshops, compiled into eight printed zines, plus [xxx][eight][online topic pages]. Work on the Toolkit also resulted in the two related datasets projects, which reflected back to inform the Toolkit. Citizen data researcher Jennifer Gabrys says toolkits "provide instructions not just for assembly and use but also for attending to the social and political ramifications of digital devices." She says they are spaces of "instruction, contingency, action, and alternative engagement" (Gabrys, 2019). As such, the *Intersectional AI Toolkit* hopes to provide resources and access points for engaging differently with machine learning systems in non-intimidating ways that connect different audiences.

OBJECTIVES: Through the Intersectional AI Toolkit, the need for plural perspectives on AI systems. Understand key terminology related to machine learning and to intersectionality. Share perspectives on the impact of emerging technology as it relates to you. Choose critically which AI tools and resources you will engage and how.

0.2.5 *: ***Interstitial Portals & Tactical Refusals***

- (Un)Limiting: Rebecca Horn, constraint and COVID art

[1] Ahmed says in *Living a Feminist Life* that the killjoy survival kit should contain books, things, tools, time, life, permission notes, other killjoys, humor, feelings, bodies, and your own survival kit.

- (Un)Raveling: Sonya Rapoport, fiber art and computation
- (Un)Forming: VALIE EXPORT, glitch feminism and broken machines
- (Un)Living: On Kawara, dailiness, death, and data
- (Un)Knowing: Pipilotti Rist, black boxes and trauma

Michel de Montaigne called the essay form a 'trial' or 'attempt'. These interstitial essays are attempts to speak in the nearbyness of *Coding.Care*. They are trials, in the sense of struggles, to get closer to the core of the creature by sneaking between its ribs. An oblique strategy, they glance against logical modes of critical analysis or direct address, in order to become the [thing] and probe the [thing] and interrupt the [thing] [simultaneously].

As an alternate take on "bias" (because bias cannot be "optimized" out of systems), these are bias cuts moving diagonally or diffractively across the warp and weft of the fabric of the other texts here. Cutting and sewing on the bias puts fabric in tension, making garments that take the shape of the bodies they surround. Bias cuts are ways of working with and against materials, acknowledging their limits and not resolving them to right angles. Thus, these essays sustain the research tensions of *Coding.Care*, unfurling the questions in the materials rather than folding them away.

Locating (and writing in) "a correspondence, not an assemblage," the essays join together by "living with" concepts (Ingold, 2015). They are portals to a co-existing "past-present-future" (Olufemi, 2021) for exploring our relationships with systems differently and intimately, in which "the past is not lost, however, but rather a space of potential" (Chun, 2021). [This is the threading together of nearbyness.]

> "the future is not in front of us, it is everywhere simultaneously: multidirectional, variant, spontaneous. We only have to *turn around*. Relational solidarities, even in their failure, reveal the plurality of the future-present, help us to see through the impasse, help

temporarily eschew what is stagnant, help build and then prepare to shatter the many windows of the here and now." (Olufemi, 2021)

As part of locating correspondences, the essays also use correspondance as a form, relying on epistolary address to conjure up analogue antecedents to the digital media discussed in other sections of *Coding.Care*. I read the works of several 20th century media artists as pre-responses to automated systems. Their wide range of practices — from minimalist daily rituals to queer feminist body art and performance — show how we have always—already been living in, talking about, performing with the questions amplified by automated systems, classification, and datification. Their works offer a breadth of artistic possibilities for reconsidering our relationships with computational systems — and these responses were already being established in parallel to the development of those systems. They help me reimagine how I want to respond now.

The essay form is a kind of embodied processing that moves the [corpus through the corpus], a reckoning in throat and gut that pairs bodily processing with computational processing. These interstitial essays serve, as queer scholar KJ Cerankowski writes, "to let this book be the crisis rather than about the crisis or crises, rather, a plurality of traumas and pains felt collectively and individually" (Cerankowski, 2021). Long traditions of artistic and literary outliers have maintained the need for such forms, like autotheory and lyric essay, which bridge aesthetic, personal, and political concerns. From these traditions, I am interested both in the constitutive act of form-making (as prefiguration) and in the reconsitituion of critical forms into poetic, personalized, or approachable forms (Fournier, 2021).

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